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RESEARCH ARTICLE

Multiclass Stress Detection Using Psychological Emotion Mapping and Transformer Architectures

KALYANI PENDKE^{1,2}, MAYURI DIGALWAR¹, AND CHARU GOEL¹

¹Indian Institute of Information Technology, Nagpur, Nagpur, Maharashtra 441108, India

²S. B. Jain Institute of Technology, Management and Research, Nagpur, Maharashtra 441501, India

Corresponding author: Kalyani Pendke (dtea20cse004@iiitn.ac.in)

ABSTRACT Stress is a common mental health problem that is apparent at a range of degrees of severity and in different circumstances. The conventional models to detect psychological stress are purely heuristic, mostly only binary classified, and neglect the fact that stress is complex and multi-dimensional. This paper presents a new hybrid deep learning model for multiclass stress detection by combining BERT, RoBERTa, CNN, and BiLSTM models. By leveraging the “GoEmotion dataset”, the authors relate emotions to three different stress levels: Not Stressed, Stressed, and Highly Stressed, drawing on psychological expertise. The model is also tested on two external datasets, VENT and ISEAR, preprocessed and relabeled with the same stress mapping schema for generalizability. A major feature in this work is the use of LIME, which improves interpretability by pointing out the most informative text features. The hybrid model outperforms the baselines with an accuracy and F1-score of 97.84 percent each. In addition to this, the proposed model shows strong cross-domain generalization. In the field of mental health applications, this model proves to be a strong base for an explainable and real-time model for stress detection.

INDEX TERMS Stress detection, multiclass classification, deep learning, BERT, RoBERTa, LSTM, CNN, text analytics, emotion classification.

I. INTRODUCTION

Nowadays, social media plays a crucial role in seeking mental health support [1]. Researchers have identified that anxiety, chronic stress, and emotional strain can be exhibited from the analysis of social media data [2], [3]. According to the World Health Organization (WHO), psychological stress plays a vital role in global mental health problems affecting the efficiency in professional surroundings and overall quality of life [1]. Eventually, stress detection using text has emerged as a crucial interdisciplinary field of research, drawing on domains such as mental health informatics, affective computing, and NLP [4].

Stress detection using text yields a non-intrusive, scalable solution against traditional techniques, including surveys, bio-signals monitoring, or clinical interviews [5]. Machine learning models can predict a person’s mental health with good accuracy by searching through syntactic structure,

linguistic bias, and emotional sentiment in a user’s text output [2], [5]. Yet the majority of current systems simplify the complex psychological stress problem by considering stress detection as a binary classification problem [3]. During the prediction of stress, these models use text without seeing the severity and scope of stress-based responses, which vary generally between situations and people [6].

To assess stress severity, advanced research is initiated to fill the above-mentioned gap, and has opted for multiclass classification [7]. Fine-grained classification is extremely crucial for context-aware stress monitoring, early intervention, and personalized mental health support [8]. In parallel to the above research, there has been noteworthy growth in the NLP domain with Transformer-based architectures like BERT (Bidirectional Encoder Representations from Transformers) [9] and RoBERTa (Robustly Optimized BERT Pretraining Approach) [10]. These models have produced better results in many language comprehension tasks by homogenizing the deep semantic and bidirectional connections of texts [9], [10].

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In addition, more recent developments have aimed for models directly including emotions, multitask learning frameworks, and knowledge distillation pipelines in the pursuit of building models that have greater semantic and psychological interpretability with respect to stress detection. For example, integrating emotion classification as an auxiliary task has been proven to increase stress detection accuracy, especially in a low-data or demographically skewed situation [11]. Introducing synthetic data generation has also been suggested, in this case with the inclusion of Large Language Models (LLMs), to be used to alleviate domain-specific data restrictions and increase the scope for emotion-aware AI in mental health [12].

However, the building blocks need to be assembled into an end-to-end stress detection pipeline, which makes a single architecture that can compactly handle semantic modeling together with temporal tracking and pattern extraction for multiclass stress detection remains limited [7], [8]. Additionally, although some models perform well on cleaned corpora, typically they are not able to generalize well across a range of uncured corpora corresponding to various linguistic styles and emotion distributions [5], [6].

The important contributions of the proposed research are:

- * Psychological-based emotion mapping: The presented work connects emotions with stress labels, directed by psychological experts, to ensure whether the stress interpretations are clinically relevant and consistent.

- * Classification at multiple levels: The proposed model analyzes the detailed mental health parameters by detecting various levels of stress.

- * Design of Model: A novel model is designed to better understand and classify stress in context by using BERT, RoBERTa, CNNs, and BiLSTMs.

- * Generalization: The presented model generalizes well on two different text datasets, i.e., ISEAR and VENT, sustaining strong performance and showing the fact that it can give good results across multiple areas.

In summary, the proposed model introduces an end-to-end framework for text-based multiclass stress detection, inclusive of current developments in deep learning and psychological modeling. It reconciles between technical developments in NLP and the societal need for explainable mental health monitoring tools.

II. RELATED WORK

Stress detection through text data is a novel and vital research field in NLP. It has been driven by the high use of social media for human communication.

During the early days of research for text-based stress detection, traditional machine learning algorithms and lexicon-based methods were extensively used. The authors in [2], [3], [4], and [5] applied attributes like emotion lexicons, such as LIWC, POS tags, and TF-IDF scores, and required contextual knowledge for validation among various text datasets.

The initiation of deep learning models such as LSTM, BiLSTM architectures, and RNNs changed the above situation [6], [7]. These models are more successful in collecting the temporal dependencies in text data [8], [18]. Another method is to use CNNs separately and in combination with RNN-based models. They become popular by their ability to extract local information from small pieces of text [6], [7].

The Transformer-based architectures, especially BERT (Bidirectional Encoder Representations from Transformers) [9] and its upgraded version RoBERTa (Robustly Optimized BERT Pretraining Approach), bring a major improvement in the field of text-based stress detection [10]. These models use various types of attention mechanisms to improve the bidirectional understanding of text. Numerous studies have conveyed that the RoBERTa model outperforms other deep learning models in the emotion classification task, especially for large-scale datasets [1], [15].

Some hybrid architectures have been shown to outperform standalone methods in the detection of stress and mental health issues. The hybrid model uses BERT with CNN for the detection of stress, and the author uses BiLSTM with BERT in another work [12], [13]. The authors in [14] used RoBERTa with GRU layers for the accurate detection of mental distress. They can give you great results by combining global semantics and local syntactic features, which is commonly very difficult to achieve with a simple single model. The research work given by the authors in [11] shows that BERT outperformed traditional and RNN-based models on the mental health-related tweet dataset.

Another line of work has focused on the lack of dataset diversity and training set generalization. Emo Pillars: A Synthetic data generation framework introduced by the authors in [4] uses Language Models like GPT-4, SST-2, and Mistral-7b for augmenting the data. The authors in [3] recently focused on the critical requirement of emotionally enriched models that are also demographically informed, enhancing multi-task learning and emotion-infused embeddings for generalization under resource constraints.

Recently, ensemble and calibration methods have been developed to enhance the reliability of models and to mitigate overfitting. The authors in [21] proposed the ensemble models for multiclass categorization of mental health. The authors in [20] proposed calibration methods for transformer models used in stress detection. The authors in [22] propose multimodal approaches. Real-world stress monitoring was also realized by the fusion of text with physiological and biometric signals in [23]. The authors in [19] describe the latest advancements and make a case for integrative hybrid and multimodal approaches.

According to recent literature, transformer-based architectures play a crucial role in stress and mental health detection from text data. Various studies extend conventional transformer models by combining emotional features and hierarchical contextual modeling to effectively collect indirect emotional indicators rooted in social media and communicative text [31], [32], [33], [34], [35], [36], [38], [39].

Moreover, research that uses large-scale emotion datasets signifies improved generalization and robustness, specifically in multiclass mental health classification, highlighting the prominence of emotional intensity over binary classification [32], [34], [36], [39].

Simultaneously, recent advancement focuses on hybrid architectures that integrate transformers with convolutional or sequential components to collect linguistic [32], [33], [35], [37], [38]. For emotionally prominent text, the transformers perform better at contextual semantic encoding, recurrent layers such as BiLSTM enhance temporal dependency modeling, and CNN-based models improve local feature extraction. Furthermore, research reports significant improvement in performance parameters [32], [35], [38], [39].

The literature points toward Transformer-based architectures as the best choice, and hybrid models that include sequential encoders, attention mechanisms, and CNNs are very important. Nevertheless, major limitations persist: there is a lack of systems for multiclass stress detection grounded in psychological theory; very few systems are available with high performance and explainability. Proposed system bridges these gaps by generating stress classes from emotion-to-stress mapping, using a novel hybrid architecture BBRC (BERT + BiLSTM+ RoBERTa + CNN), validation over two text datasets VENT and ISEAR, and interpretable results through LIME. The proposed model takes a holistic approach to simultaneously be technically robust and clinically interpretable.

III. METHODOLOGY

The following section gives details of the proposed method, system architecture, and mathematical formulation of the proposed model.

The authors have developed a hybrid model and compared three deep learning models to efficiently capture and classify varying stress levels in text. The proposed hybrid model merges the potential of transformer-based embeddings (BERT and RoBERTa), sequential modeling (LSTM and BiLSTM), and convolutional layers to extract diverse linguistic characteristics.

All the models are trained on the “GoEmotion dataset”, where each text is mapped to a stress label based on underlying emotions: “joy” representing Not Stressed, “fear” and “anger” for Stressed, and “sadness” for Highly Stressed. This derived dataset was curated and reviewed in consultation with a psychologist. All preprocessing steps, such as tokenization and text normalization, are performed to clean the text data.

The RoL model has used RoBERTa as a contextual embedding creator. The output embeddings are then passed through a unidirectional LSTM layer, which captures sequential dependencies in the token representations. A dense layer using the softmax activation function has carried out the final classification of three stress levels.

The RoBi model is a combination of the RoBERTa algorithm and Bidirectional LSTM (BiLSTM). The RoBERTa

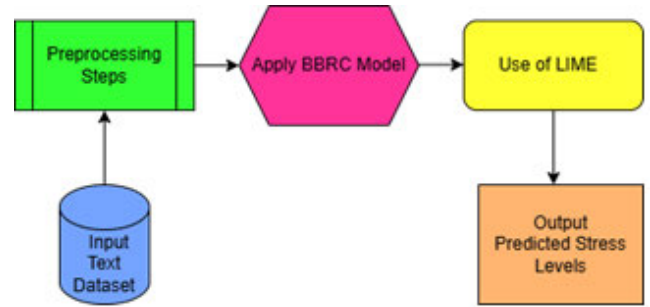


FIGURE 1. Overview of the proposed model.

algorithm gives contextual embeddings, and the BiLSTM identifies the context of tokens in forward and backward directions. This advancement enhances the model’s capability to learn linguistic elements related to stress.

The BERT embeddings and the 1D Convolutional layer are combined to form the BC model. In this model, the BERT embeddings are used as input features, and to identify local patterns in text data, a 1D convolutional layer is applied. Emotional transitions, which are identified by the short-term word co-occurrence sequence, are recognized by this model. The final class probabilities are obtained when the CNN output is passed through dense layers.

Figure 1 portrays the overview, and Figure 2 gives a detailed working of the proposed model. It uses the features of four different deep learning models. BERT and RoBERTa are used for understanding the context. To understand the sequential emotion flow, BiLSTM is used, and for local pattern recognition, CNN is applied. The above components are trained using a categorical cross-entropy loss function to predict one of the three stress classes. The first step is to learn contextual embeddings, for which the tokenized text is passed through the BERT and RoBERTa encoders. The outputs of the first step are combined and forwarded to the BiLSTM and CNN modules in parallel. To give final class predictions, the results are further merged and given to a dense layer with a softmax activation function.

LIME is further used to enhance the model’s interpretability. This explainability technique contributes to learn the predictions of closed box models. During testing, LIME is combined with the final trained model to envision and infer decision rationales. This enhancement identifies the crucial lexical features related to distinct levels of stress. It is also ensured that the model hasn’t worked on irrelevant clues.

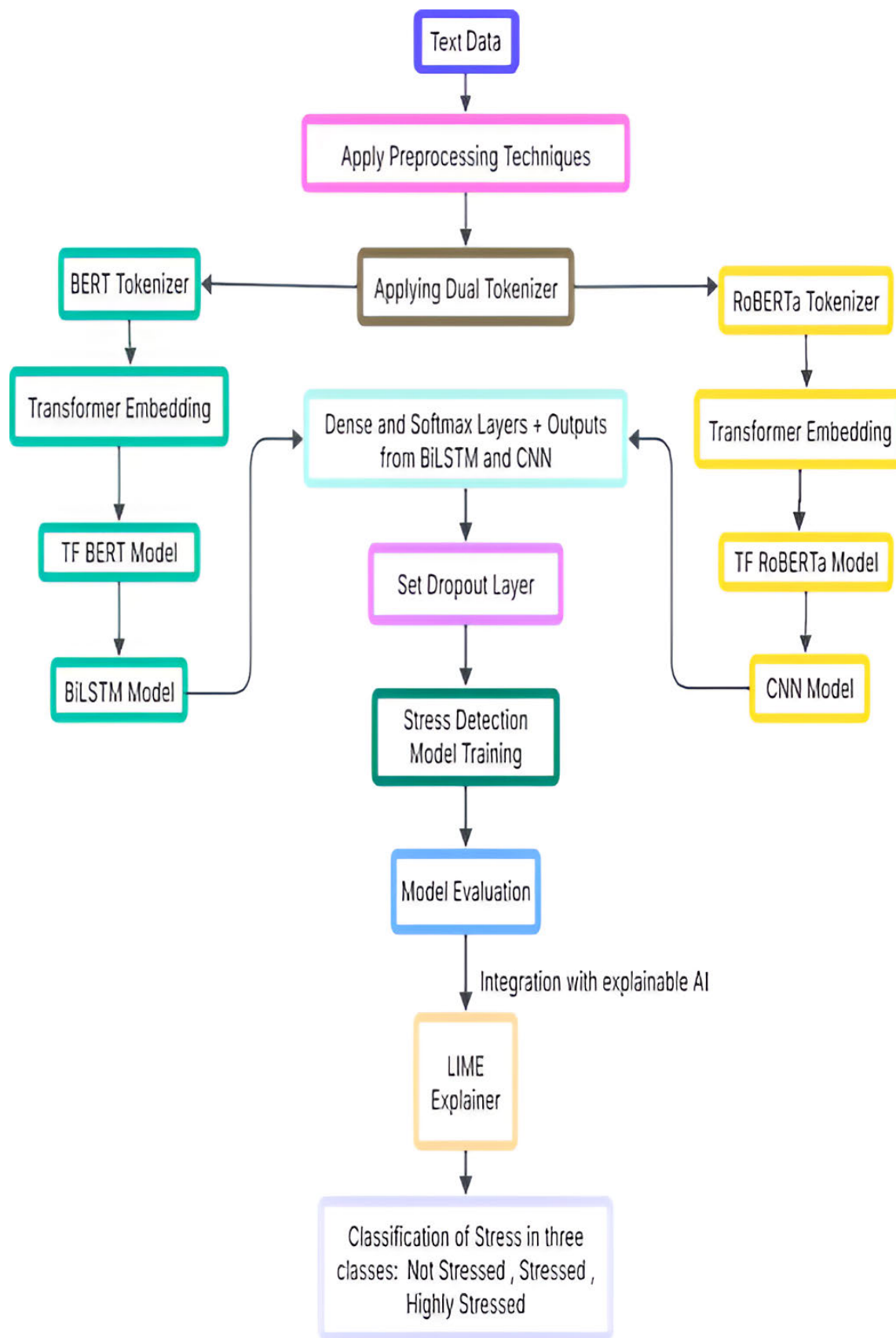
A. MATHEMATICAL FORMULATION OF THE PROPOSED MODEL

The presented model combines BERT, BiLSTM, RoBERTa, and CNN, each contributing sequence modeling and distinct representational capabilities.

A preprocessed text sequence is given as input to the model (X), which is represented as follows,

$$X = \{w_1, w_2, \dots, w_T\} \quad (1)$$

where, T represents the number of tokens.

**FIGURE 2.** Detailed Working of Proposed Model.

The pretrained tokenizers of BERT and RoBERTa tokenize and encode the input text, generating the corresponding embedding matrices (EBERT and ERoBERTa), which are given as follows,

$$E^{\text{BERT}} = \text{BERT}(X), \quad E^{\text{RoBERTa}} = \text{RoBERTa}(X) \quad (2)$$

where, each $E \in \mathbb{R}^{T \times d}$ with d being the hidden dimension size.

The concatenation of these embeddings is performed E_{concat} , which is shown in the following equation,

$$E_{\text{concat}} = [E^{\text{BERT}}; E^{\text{RoBERTa}}] \in \mathbb{R}^{T \times 2d} \quad (3)$$

The results of the above concatenation are further processed through BiLSTM to collect both forward and backward contextual dependencies, which are represented in the following equations,

$$\vec{h}_t, \overleftarrow{h}_t = \text{LSTM}(E_{\text{concat}}) \quad (4)$$

$$H_t^{\text{BiLSTM}} = [\vec{h}_t; \overleftarrow{h}_t] \in \mathbb{R}^{T \times 2h} \quad (5)$$

where, h is the hidden size of the LSTM and H_t^{BiLSTM} represents the contextualized hidden states.

The same embedding E_{concat} is parallelly transited through a one-dimensional CNN with ReLU activation function (C), as shown below,

$$C = \text{ReLU}(\text{Conv1D}(E_{\text{concat}})) \quad (6)$$

Max pooling is used to extract important features (Cpooled),

$$C_{\text{pooled}} = \text{MaxPool}(C) \quad (7)$$

In the next step, the concatenation of the outputs of the BiLSTM and CNN modules is performed (F),

$$F = [\text{Flatten}(H_t^{\text{BiLSTM}}); \text{Flatten}(C_{\text{pooled}})] \quad (8)$$

The vector F is passed through a fully connected dense layer, following this, a softmax classifier is used to give class probabilities for the three stress levels:

$$z = W_f F + b_f \quad (9)$$

$$\hat{y} = \text{Softmax}(z), \quad \hat{y} \in \mathbb{R}^3 \quad (10)$$

The optimization of the model is performed using the categorical cross-entropy loss function (L):

$$L(y, \hat{y}) = - \sum_{i=1}^3 y_i \log(\hat{y}_i) \quad (11)$$

where, $y \in \{0, 1\}^3$ is the one-hot encoded true label. The Adam optimizer is used to minimize this loss.

IV. RESULTS

The proposed model is systematically studied in the current section, which includes experimental setup, the dataset, experimental results, discussion, and cross-dataset validation.

The first subsection gives information about the text datasets used for experiments. The second subsection talks about the hyperparameter tuning and training procedures applied to attain good performance. The third subsection presents the results of experiments, and it also gives the extended epochs analysis. The last subsection discusses the applicability of the models in real-world scenarios and how the models generalize to unseen data.

A. DATASET

The “GoEmotion dataset” [24] that Google Research created is used in this work to conduct multiclass stress detection. This dataset consists of 58,101 Reddit comments annotated with emotion categories, such as joy, sadness, fear, anger, love, and surprise. The GoEmotion dataset does not have explicit annotations for stress levels. Accordingly, the most significant role of this research is to classify new data by utilizing the emotion-to-stress mapping and to ground these labels in stress-based classification.

Based on an elaborate literature study and expert consultation with a professional psychologist, six emotions were chosen from the GoEmotion dataset. The category “Not Stressed” was correlated with the emotions of joy and love. Within these hypotheses, surprise, anger, and fear are the precursors to moderate levels of psychological stress, so they created a “Stressed” category. When considering all of the classes that were mapped together with a true emotion, the authors found that only the “Highly Stressed” class was always mapped to sadness or any other deep emotional distress. The resulting emotion-to-stress mapping is in line with millennia of research in affective computing and psychological modeling, proving the clinical relevance of the newly derived dataset.

Through this mapping, the final label distribution resulted in 8,402 samples for the “Not Stressed” class, of which 6,761 are joy, and 1,641 are love. Of main interest here was the “Stressed” psychological state, which contained 5,801 samples: comprising 719 surprise, 2,709 anger, and 2,373 fear instances. 5,797 sadness-labeled samples belonged to the “Highly Stressed” category.

Overcoming the imbalance and ensuring equitable training was achieved by using an undersampling technique to create a balanced dataset. All classes were subsampled to around 7,119 rows in total, which is fairly balanced and separated among train, validation, and test sets. The training set contained 4,983 samples, and there were 1,661 instances in each class. For each class number, 356 samples were in the validation, and the test set contains 1,068 samples. The authors define a class-balanced configuration that sets all the stress classes to have the same representation during training

and evaluation. Such a config makes it less likely for a model to be biased towards the most frequent classes.

The quality of the dataset is improved by applying preprocessing techniques. Those were removing some null entries, empty strings, and sentences made only of non-informative or meaningless symbols. Input texts were normalized, standardized, and tokenized using the model-specific tokenizers provided with BERT and RoBERTa architectures to ensure compatibility and optimized feature representation. Thus, the preprocessing work as a whole led to having a clean, balanced, and enriched in semantics dataset for exposing training and evaluation of multiclass stress detection algorithms.

The proposed model uses pretrained transformer encoders trained on a huge number of tokens. Thereby, the model builds on current linguistic knowledge and fine-tunes high-level semantic features, minimizing the requirement for substantial task-specific data.

B. EXPERIMENTAL SETUP

This section details the model configuration used for multiclass stress identification. The authors run experiments in a controlled environment to be able to do a fair comparison between their proposed model and the baseline models.

To keep the experiment consistent, all four models, RoL, RoBi, BC, and the proposed hybrid model, were trained using the same hyperparameters. The models were trained for 21 epochs with a batch size of 32, using the Adam optimizer. The authors chose a learning rate of $5e-5$ after trying a few rates. The dataset was split into training (80 percent), testing (10 percent), and validation (10 percent). Stratified sampling was used to ensure stress classes were equally distributed among the splits.

To improve generalizations and avoid overfitting, two callbacks were used: EarlyStopping (with a patience of 5 epochs) and ReduceLROnPlateau (patience of 3 epochs). A sparse categorical cross-entropy loss function was selected for the multiclass classification. Furthermore, to increase sensitivity, primarily for Highly Stressed samples, class weights were calculated from training data. The evaluation metrics used are Precision (P), Recall (R), Accuracy (A), and F1-score. It is important to note that the authors have used these hyperparameters as constants in all four model configurations to ensure a fair comparison. All experiments were implemented and run on the Kaggle platform with publicly available Jupyter notebooks. The list of key hyperparameters is given in Table 1.

C. EXPERIMENTAL RESULTS

The following section gives a detailed investigation of the obtained results, including results obtained using twenty-one epochs and an extended epochs analysis.

1) RESULTS WITH TWENTY-ONE EPOCHS

The following subsection details the results obtained by running all four models for twenty-one epochs. It gives

TABLE 1. List of key hyperparameters.

Hyperparameter	Value
Tokenizer	BERT and RoBERTa
Optimizer	Adam
Batch Size	32
Loss Function	Categorical Cross-Entropy
Learning Rate	$5e-5$
Epochs	21
Evaluation Metrics	P, R, A, F1-score

TABLE 2. Results of proposed models on “GoEmotion Dataset”.

S.No	Types	Parameters	P*	R*	F1-Score	A*	Imp*
1	RoL	Not Stressed	64.46	60.28	62.30	68.54	—
		Stressed	78.51	77.18	77.84		
		Highly Stressed	63.02	68.17	65.49		
		Macro Avg	68.66	68.54	68.54		
2	RoBi	Not Stressed	80.78	87.61	84.05	85.73	—
		Stressed	91.27	85.35	88.21		
		Highly Stressed	85.92	84.23	85.06		
		Macro Avg	85.99	85.73	85.78		
3	BC	Not Stressed	93.50	93.24	93.37	94.18	—
		Stressed	95.95	93.52	94.72		
		Highly Stressed	93.15	95.77	94.44		
		Macro Avg	94.20	94.18	94.18		
4	BBRC	Not Stressed	97.46	97.18	97.32	97.84	+3.88
		Stressed	97.51	99.44	98.47		
		Highly Stressed	98.57	96.90	97.73		
		Macro Avg	97.85	97.84	97.84		

*P (Precision), R (Recall), A (Accuracy) and Imp (Percentage Improvement)

model performance comparison, confusion matrix analysis, and graph-based analysis of the model.

a: PERFORMANCE COMPARISON OF MODELS

The classification performance of all proposed models was assessed by using standard performance metrics.

The RoL model takes advantage of contextual embeddings and sequential modeling, but it has a bottleneck in the unidirectional nature of the LSTM layer; this can lead to a misleading understanding of the sentence structure as a whole, especially with emotionally ambiguous texts. RoL model attains the value of 68.54 percent for both accuracy and macro F1-score.

The RoBi model uses the Bi-LSTM to capture past and future context, which helps the model to correctly understand emotionally complex sentences. This enhancement is especially visible in the higher recall for the Highly Stressed class, which often contains long-form expressions of fear or sadness. The RoBi model achieved the value of 85.73 percent for accuracy and 85.78 percent for macro F1-score.

For the BC model, the authors combine BERT with CNN. BERT is used for its strong semantic representation, and CNN for its capability to collect local n-gram sequences. This model has the ability to distinguish neutral from emotionally negative content. It shows an accuracy of 94.18 percent, and a macro F1 score is also 94.18 percent.

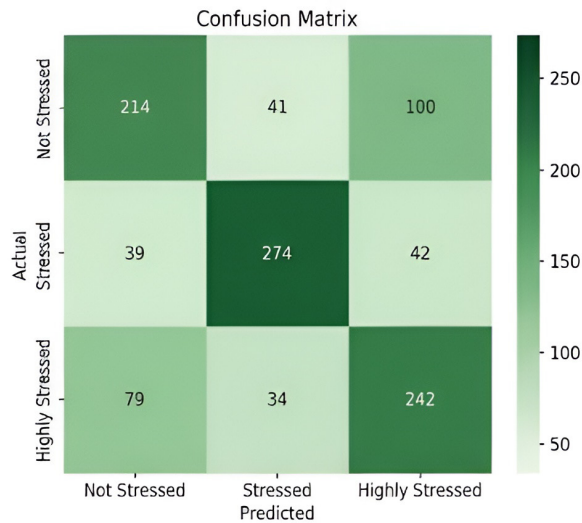


FIGURE 3. Confusion matrix of RoL.

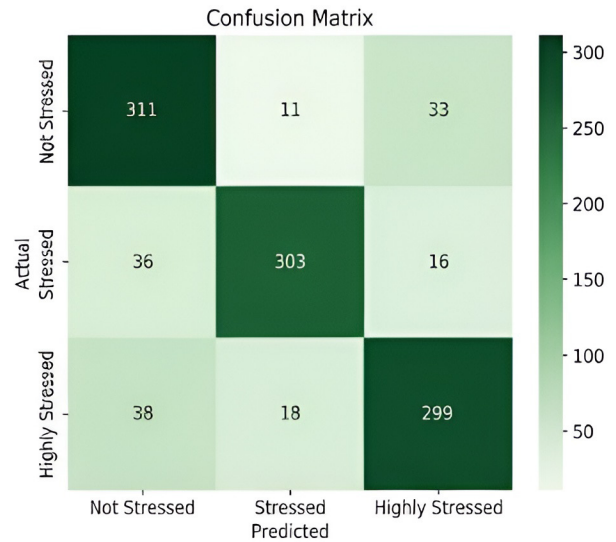


FIGURE 4. Confusion matrix of RoBi.

Among all the above models, the most noteworthy results were observed in the proposed hybrid model, which integrates BERT, BiLSTM, RoBERTa, and CNN. This model attained an accuracy and F1-score of 97.84 percent. The hybrid structure enables the model to simultaneously use deep contextual semantics obtained from BERT and RoBERTa, sequential understanding generated by using BiLSTM, and localized phrase detection through CNN. This architecture allows good generalization among all stress categories. Table 2 summarizes the performance results of all four models.

The success of the proposed BBRC model is emphasized by giving the performance gains over the highest baseline models. As mentioned in Table 2, the BBRC model attains a relative enhancement of 3.88 percent in accuracy over the BC model, the best-performing baseline for the GoEmotion dataset. This signifies the advantage of using dual transformers with parallel CNN and BiLSTM modules.

b: ANALYSIS OF CONFUSION MATRICES

The confusion matrices of all four models represent their capacity to classify the text in the above-mentioned three stress categories.

The confusion matrix of the RoL model, shown in Figure 3, shows that a considerable number of Highly Stressed examples were misclassified as Stressed. This confusion displays that it is difficult for the model to capture fine-grained emotional differences between fear, sadness, and surprise. The clear and dissimilar nature of positive emotions like joy and love helps the model to correctly classify the Not Stressed class.

Figure 4 shows the confusion matrix for the RoBi model. The diagonal dominance indicates that the model effectively differentiates between all three stress levels. The confusion

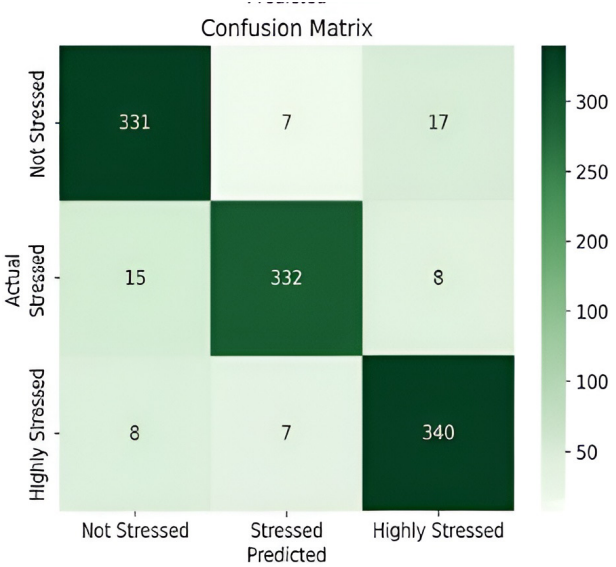


FIGURE 5. Confusion matrix of BC.

matrix also shows that the small amount of overlap lies between the Stressed and Highly Stressed classes.

Figure 5 represents the confusion matrix for the BC model, which shows that most predictions fall along the diagonal. This model correctly separates Not Stressed text from both stress-persuading categories and reveals high precision in identifying emotionally intense text.

Figure 6 gives the confusion matrix for the proposed hybrid model, which is the most accurate of all four models. It shows that the proposed hybrid model can differentiate even the most slightly different emotional categories. The small number of misclassifications can be credited to ambiguous texts or those comprising overlapping emotions. However, the matrix authenticates the robustness of the proposed hybrid model.

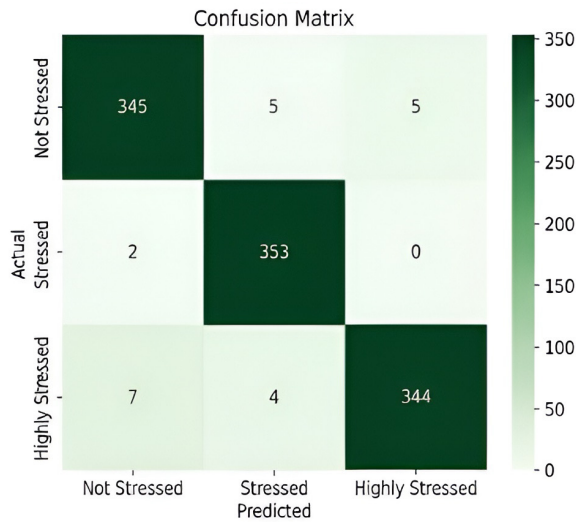


FIGURE 6. Confusion matrix of BBRC.

c: ANALYSIS OF MODEL LEARNING BEHAVIOR BASED ON GRAPH

Each model's convergence behavior, learning efficiency, and generalization capability can be learnt by in-depth examination of training and validation accuracy and loss curves.

Figure 7 illustrates the RoL model's performance. It displays that the training accuracy increases slowly and plateaus after 18 epochs. The validation accuracy remains relatively low, signifying that the model fails to collect complex emotional patterns in the validation data. The corresponding loss curves illustrate a steady reduction in training loss, while the validation loss plateaus at a higher level, signifying partial generalization ability.

Figure 8 shows the RoBi model's performance. The training and validation accuracy curves are aligned from the early epochs, and both progressively enhance over time, indicating a well-regularized model. This model evades overfitting and learns the stress levels decision boundaries effectively, as validation loss and training loss are approximately similar.

The effective work of the BC model is shown in Figure 9. In the first 10 epochs, the early flattening of the loss curve is observed, and the accuracy increases sharply, while after 90 percent, it becomes consistent. This proves that the model gives better results with relatively fewer epochs. The acceleration of convergence is a result of the addition of CNN layers in the model, which helps in collecting better local patterns in early training.

The proposed BBRC model's performance is envisioned in Figure 10. This model outperforms other configurations in terms of training dynamics. By the 15th epoch, the training and validation accuracy rise to 95 percent, whereas at around the 21st epoch, it reaches 98 percent. A reliable and effective learning process is illustrated by the loss curve, as it shows a steep drop and then scales moderately. In this model, the validation performance keeps track of the training

TABLE 3. Extended epoch evaluation and convergence analysis.

Model	Parameters	Epoch (10)	Epoch (20)	Epoch (40)	Epoch (70)	Epoch (100)
RoL	Accuracy (%)	61.60	70.70	76.06	86.76	88.92
	F1-Score (%)	61.95	70.56	76.10	86.79	88.92
RoBi	Accuracy (%)	74.65	83.47	93.62	98.12	97.93
	F1-Score (%)	74.57	83.38	93.64	98.12	97.93
BC	Accuracy (%)	81.69	93.15	98.59	99.15	98.87
	F1-Score (%)	81.55	93.15	98.59	99.15	98.87
BBRC	Accuracy (%)	84.51	97.75	98.22	98.12	98.03
	F1-Score (%)	84.29	97.75	98.21	98.12	98.03

performance, which results in no overfitting. This hybrid architecture is scalable and efficient, as shown by the steady convergence.

2) EXTENDED EPOCHS ANALYSIS

To compare the behavior and stability of training, the extended epoch analysis is performed, where each model is trained using five different numbers of epochs: 10, 20, 40, 70, and 100. This analysis enables us to understand the relationship between model performance that changes with longer training, and to find out the stage where each model converges stably without overfitting. As presented in Table 3, all models show a gradual increase in both accuracy and F1-score with the increased number of epochs. The RoBi model shows more stable improvement, achieving 93.6 percent accuracy at the 40th epoch and maintaining an accuracy of around 98 percent in longer training, indicating better generalization and convergence.

In the BC model, the accuracy reached 93.15 percent by 20 epochs and exceeded 98 percent by 40 epochs. After that, its performance became stable, showing that it converged effectively within fewer epochs. The accuracy of the RoL model increases from 61.6 percent to 88.9 percent as the number of epochs increases, but after 70 epochs, the results start fluctuating, indicating the beginning of overfitting. The newly developed hybrid model BBRC attains high accuracy at an early stage, recording 97.75 percent at 20 epochs and remaining close to 98 percent up to 100 epochs. The consistent performance noticed in later epochs shows that the hybrid model learns efficiently and sustains generalization without overfitting during longer training.

D. DISCUSSION

This section examines the obtained results in detail by discussing the effectiveness of the stress detection model, ablation study, statistical significance analysis, and the explainability using the LIME technique.

1) EFFECTIVENESS OF STRESS DETECTION MODEL

The usefulness of the presented hybrid model is validated by comparing the obtained results with various current state-of-the-art models. Table 4 gives a summary of the obtained results, where all the models are evaluated on the GoEmotion dataset.

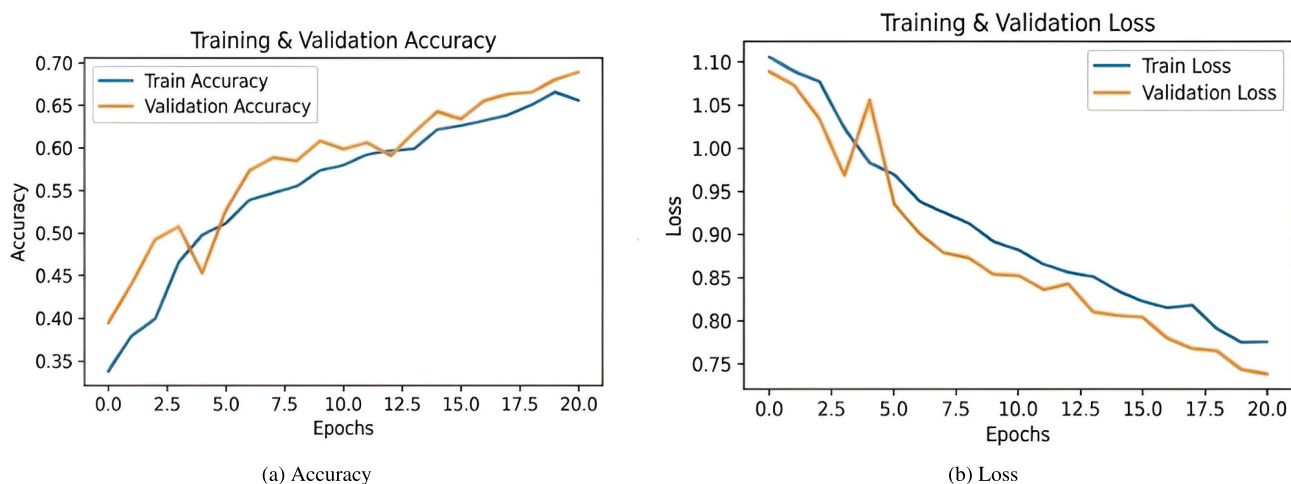


FIGURE 7. Performance Graphs of the RoL model.

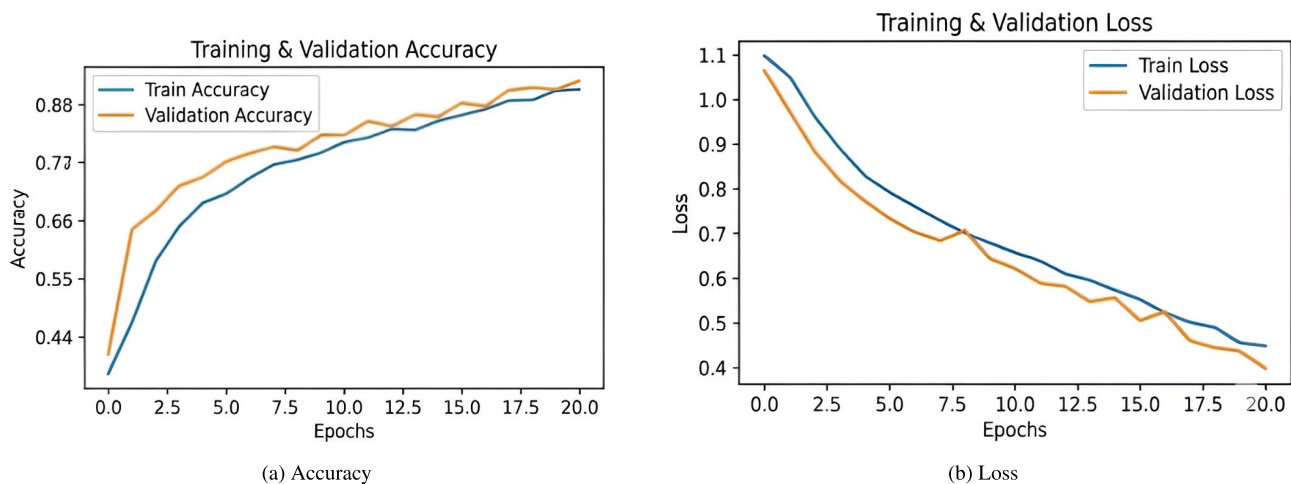


FIGURE 8. Performance Graphs of the RoBi model.

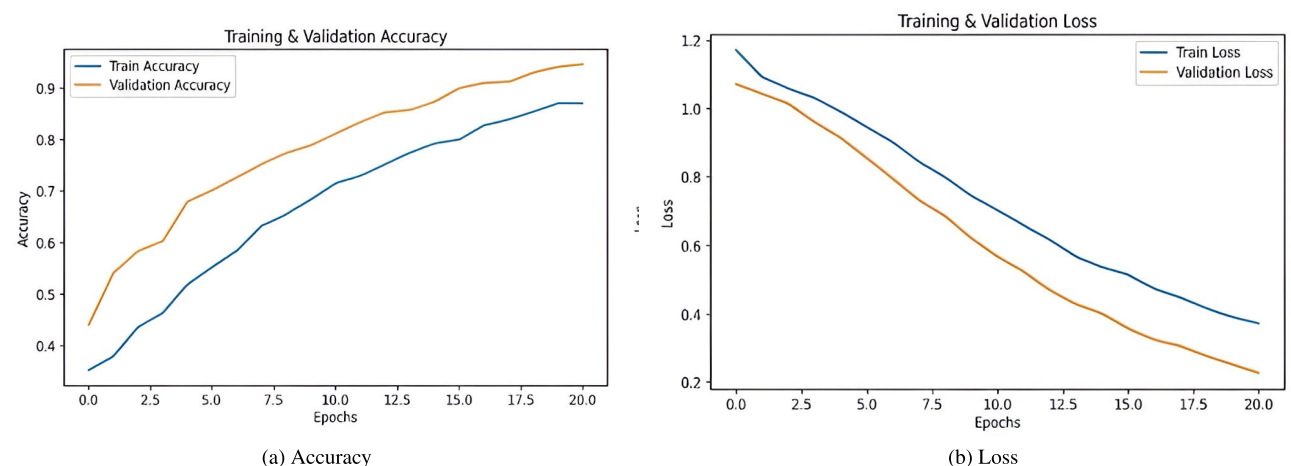


FIGURE 9. Performance Graphs of the BC model.

The results in [2] show that the RoBERTa model attained the highest value of a weighted F1-score of 84.58 percent. This work lags the stress-specific classification setup

proposed in the present study. For stress detection, the authors in [3] developed emotion-infused models, where interpretability is improved by using multi-task learning.

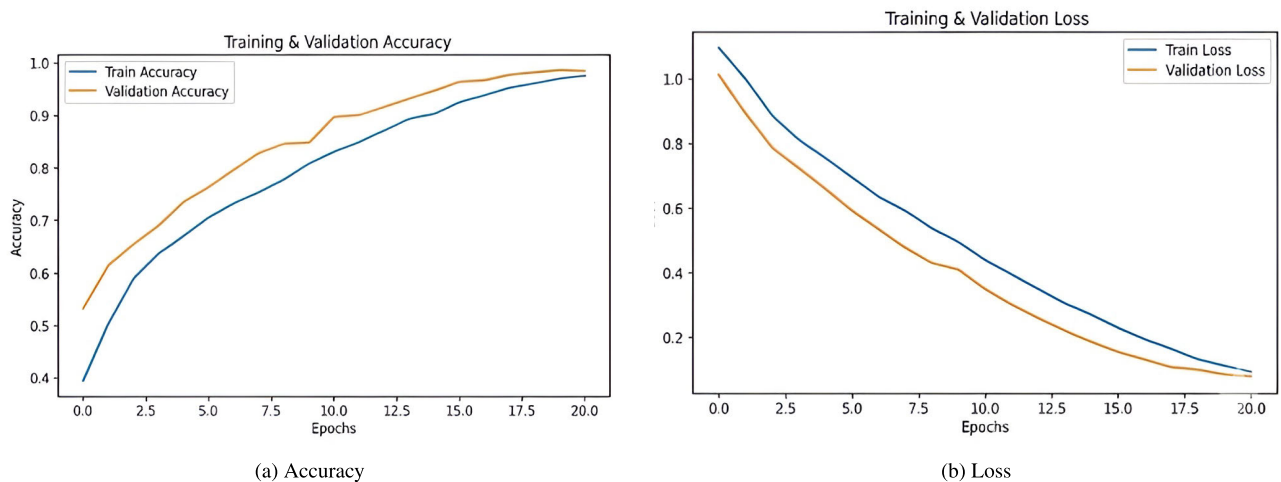


FIGURE 10. Performance Graphs of the BBRC model.

TABLE 4. Performance comparison of various models.

Sr. No	Algorithm	A* (%)	F1-Score (Weg) (%)	Imp* (%)
1	RoBERTa [2]	—	84.58	—
2	Multitask Model [3]	81.07	80.24	—
3	MentalRoBERTa [4]	78.53	74.52	—
4	RoBERTaOrig-fine-tuned [5]	—	55.00	—
5	RAMHA [32]	88.00	—	—
6	Stacked LSTM [37]	—	50.00	—
7	BBRC (Proposed Approach)	97.84	97.84	+11.18

*A (Accuracy) and Imp (Percentage Improvement)

This work falls behind performance parameters related to multiclass stress classification.

The authors in [4] focus on the performance enhancement in low-resource environments by delving multi-task models for stress detection. The Emo Pillars, discussed by the authors in [5], is a knowledge simplification method that uses data produced from various types of LLMs. They discuss the method that enhances domain adaptation and contextual learning, but this method is less efficient than lightweight design and efficient models.

Table 4 shows that the latest models, such as RAMHA [32] and stacked LSTM [37], exhibit the efficiency of contextual and sequential modeling for emotion and stress detection from social media text. Although performance differs over datasets and architectures. However, the proposed BBRC model attains 97.84 percent accuracy and a weighted F1-score on the GoEmotion dataset, signifying a clear performance enhancement.

The success of the proposed BBRC model is emphasized by giving the performance gains over the highest baseline models. As mentioned in Table 4 regarding the latest model, the proposed BBRC model attains a relative enhancement of 11.18 percent in accuracy over the best-performing baseline model, RAMHA, mentioned in [32].

TABLE 5. Findings of ablation study.

Model	P* (%)	R* (%)	F1-score (%)	A* (%)
BERT	44.22	43.00	42.53	43.00
RoBERTa	11.11	33.33	16.67	33.33
BR	46.14	44.98	44.50	44.98
BBi	67.22	66.57	66.44	66.57
BC	98.50	98.50	98.50	98.50
RoBi	90.68	89.95	89.97	89.95
RoC	93.91	93.80	93.80	93.80
BRoBi	80.21	79.72	79.56	79.72
BRoC	93.74	93.71	93.71	93.71
RoBiC	90.93	90.80	90.77	90.80
BBiC	87.32	86.95	86.87	86.95
BBRC (Proposed Approach)	98.05	98.03	98.02	98.03

*P (Precision), R (Recall), A (Accuracy)

2) ABLATION STUDY

The authors performed a detailed ablation study to explain the architectural design of the proposed BBRC model and to observe the separate role of each component. Table 5 gives the results of all models that were trained under similar experimental settings for 34 epochs to ensure a fair comparison.

Table 5 results reveal that single transformer models underperform for the multiclass stress classification task, with BERT and RoBERTa attaining F1-scores of 42.53 percent and 16.67 percent, respectively. This specifies that contextual embeddings alone are inadequate to collect subtle stress cues in text. The BR (BERT + RoBERTa) model also fails to show enhancement, attaining only 44.50 percent F1-score, indicating that a simple merger of pretrained encoders does not give assurance for complementary learning.

The RoBi (RoBERTa + BiLSTM) model introduces sequential modeling, which notably enhances performance with an F1-score of 89.97 percent. But the significant improvements are observed after the addition of convolutional layers. The BC (BERT + CNN) model gives

TABLE 6. Accuracy and F1-Score (Mean $\pm$ Std Dev).

Model	Accuracy (Mean $\pm$ Std)	F1-Score (Mean $\pm$ Std)	Imp* (%)
BERT	45.31 $\pm$ 1.29	45.45 $\pm$ 1.42	—
RoBERTa	33.18 $\pm$ 0.17	17.41 $\pm$ 0.12	—
BC	94.34 $\pm$ 0.83	94.27 $\pm$ 0.75	—
RoL	69.87 $\pm$ 1.91	69.66 $\pm$ 1.71	—
RoBi	85.70 $\pm$ 1.13	85.53 $\pm$ 1.22	—
BBRC	97.10 $\pm$ 0.87	97.00 $\pm$ 0.92	+2.92

*Imp (Percentage Improvement)

TABLE 7. Paired T-Test results against baseline.

Comparison	p-value (A)	p-value (F1-Score)
Baseline (BERT) vs RoBERTa	0.0005	< 0.001
Baseline (BERT) vs BC	< 0.001	< 0.001
Baseline (BERT) vs RoBi	< 0.001	< 0.001
Baseline (BERT) vs RoL	0.001	0.001
Baseline (BERT) vs BBRC	< 0.001	< 0.001

the significance of CNN-based local feature extraction in capturing stress-indicative patterns by attaining a high F1-score of 98.50 percent.

Additional experimentation with triple-component configurations reveals that very few extensions enhance performance. The BRoBi (BERT + RoBERTa + BiLSTM) model shows lesser performance with an F1-score of 79.56 percent, signifying that the combination of architecture is very important and needs careful design. The remaining combinations, BBi (BERT + BiLSTM) model, RoC (RoBERTa + CNN) model, BRoC (BERT + RoBERTa + CNN) model, RoBiC (RoBERTa + BiLSTM + CNN) model, and BBiC (BERT + BiLSTM + CNN) model, reveal less performance, signifying that a simple combination of components is inadequate. This further inspires the cautious design of the BBRC model.

The BBRC model (BERT + RoBERTa + CNN + BiLSTM) combines global contextual understanding, local pattern extraction, and sequential dependency modeling, attains an F1-score of 98.02 percent, and gives a stable architecture.

The authors performed a comprehensive ablation analysis confirming that every component in the BBRC model plays a discrete and crucial role. The experimental performance variations across various configurations show that the BBRC model is not a simple combination of existing methods, but rather a sensibly enhanced architecture where mutual learning mechanisms cooperatively contribute to higher and more reliable performance.

3) STATISTICAL SIGNIFICANCE ANALYSIS

This analysis is carried out to confirm that the performance enhancements are not due to random chance. Independent sample t-tests were used to compare the F1-scores of the proposed hybrid model against the three baseline models, RoBi, RoL, and BC, over multiple runs with four different random seeds (21, 34, 42, 54). A t-test was conducted to test

TABLE 8. Results of proposed models on the vent dataset.

S.N	Types	Parameters	P*	R*	F1	A*	Imp*
1	RoL	Not Stressed	87.20	79.00	82.90	80.33	—
		Stressed	78.30	79.40	78.85		
		Highly Stressed	76.48	82.60	79.42		
		Macro Avg	80.66	80.33	80.39		
2	RoBi	Not Stressed	92.19	85.00	88.45	86.13	—
		Stressed	84.82	83.80	84.31		
		Highly Stressed	82.20	89.60	85.74		
		Macro Avg	86.40	86.13	86.17		
3	BC	Not Stressed	95.01	87.60	91.16	89.40	—
		Stressed	88.65	89.00	88.82		
		Highly Stressed	85.29	91.60	88.33		
		Macro Avg	89.65	89.40	89.44		
4	BBRC	Not Stressed	95.79	91.00	93.33	92.60	+3.57
		Stressed	94.95	90.20	92.51		
		Highly Stressed	87.82	96.60	92.00		
		Macro Avg	92.85	92.60	92.62		

*P (Precision), R (Recall), A (Accuracy) and Imp (Percentage Improvement)

the null hypothesis with $\alpha = 0.05$. The obtained p-values were all below 0.05, substantiating the proposed model's statistically significant performance. The variability arising from random initialization and stochastic training behavior is a result of this approach. Table 6 gives the values of Mean and Standard deviation of six models, while Table 7 shows the paired T-test values against the baseline models.

4) EXPLAINABILITY USING LIME

The proposed model uses LIME to increase its interpretability. This technique provides local, human-understandable explanations for individual predictions made by complex closed box models. This enables researchers to recognize the most influential input features in the final classification, giving insights into the decision-making process of the model.

The presented work uses LIME to study the output of the final hybrid model on demonstrative samples from three different stress classes. Figure 11 shows the detailed working of LIME for an input text sample. The presented model correctly identified the category for this text as “Not Stressed” with a confidence score of 56 percent.

5) GENERALIZATION AND CROSS-DATASET VALIDATION

The cross-dataset validation was performed to assess the generalization capability and robustness of the presented model. The datasets used for validation are VENT and ISEAR, which were carefully preprocessed, filtered, and mapped to align with the three stress categories Not Stressed, Stressed, and Highly Stressed, formerly defined in association with psychological experts.

α : VALIDATION ON VENT DATASET

The VENT dataset contains approximately 475,980 user-generated posts. As the proposed model focuses on the six

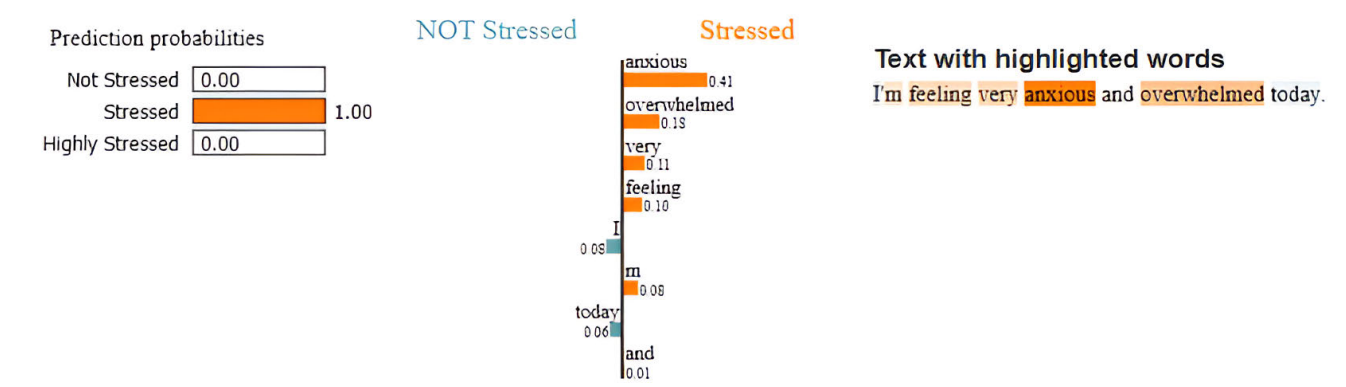


FIGURE 11. LIME explanation highlighting word-level contributions.

emotions used in the GoEmotion experiment, the current task also considers the posts expressing the same six emotions.

To ensure data quality, extensive preprocessing, including coherence filtering, noise reduction, and null value removal, was used. To create a balanced dataset, undersampling was carried out. A representative subset of 9,000 samples was selected to reduce computational load and avoid overfitting, which is further divided into training (6,000), test (1,500), and validation (1,500) sets.

The training of the four models was performed for twenty-one epochs and evaluated on this balanced test set. Table 8 shows that the hybrid model BBRC attained the best performance with an accuracy of 92.60 percent and a macro F1-score of 92.62 percent, indicating strong generalization capacity. The BC model achieved an F1-score of 89.44 percent, and the RoBi model, with an F1-score of 86.17 percent, also worked effectively. Though the RoL model underperformed compared to the other models.

b: VALIDATION ON ISEAR DATASET

The ISEAR dataset contains self-reported and psychologically elicited emotional descriptions obtained through a standardized survey. Its language is comparatively formal and undiversified in comparison with large-scale social media data. The ISEAR dataset contains text, labeled with seven basic emotions, out of which only four emotions were mapped with the proposed stress classification task.

The Joy emotion was mapped to the Not Stress class containing 1094 samples. Two emotions, anger and fear, were mapped to the Stress class with 2191 samples. The sadness emotion was mapped to the Highly Stress class containing 1096 samples. To ensure consistency, the same preprocessing steps as in the VENT dataset were performed.

The training of the four models was performed for twenty-one epochs and evaluated on this balanced test set. Table 9 shows that the hybrid model BBRC attained the best performance with an accuracy of 81.30 percent and a macro F1-score of 81.20 percent, indicating strong generalization capacity. The BC model achieved an F1-score

TABLE 9. Results of proposed models on the ISEAR Dataset.

S.N	Types	Parameters	P*	R*	F1	A*	Imp*
1	RoL	Not Stressed	81.56	89.02	85.13	76.02	-
		Stressed	77.61	63.41	69.80		
		Highly Stressed	69.27	75.61	72.30		
		Macro Avg	76.15	76.02	75.74		
2	RoBi	Not Stressed	89.81	85.98	87.85	80.89	-
		Stressed	76.22	76.22	76.22		
		Highly Stressed	77.19	80.49	78.81		
		Macro Avg	81.07	80.89	80.96		
3	BC	Not Stressed	84.53	93.29	88.70	80.28	-
		Stressed	80.99	70.12	75.16		
		Highly Stressed	75.15	77.44	76.28		
		Macro Avg	80.22	80.28	80.05		
4	BBRC	Not Stressed	85.63	90.85	88.17	81.30	+1.27
		Stressed	81.88	74.39	77.96		
		Highly Stressed	76.33	78.66	77.48		
		Macro Avg	81.28	81.30	81.20		

*P (Precision), R (Recall), A (Accuracy) and Imp (Percentage Improvement)

of 80.05 percent, and the RoBi model, with an F1-score of 80.96 percent, also worked effectively. Though the RoL model underperformed compared to the other models.

V. CONCLUSION

This study proposes a hybrid deep learning framework integrating psychological emotion mapping alongside sophisticated NLP models, BERT, RoBERTa, CNN, and BiLSTM for multiclass stress detection. The model’s performance is evaluated on the GoEmotion dataset. The validation is performed with the help of the VENT and ISEAR datasets to achieve the generalizability. The model’s understanding is improved with the help of LIME, which provides transparency in prediction. This improvement plays a crucial role in the fields related to mental health.

Future work can use a larger dataset than the one used in the proposed work to improve the model’s generalization. In spite of the impressive performance of the proposed BBRC model, some challenges should be acknowledged for real-world deployment. The BBRC model is mainly

evaluated on English-language datasets, and its output may be influenced when exposed to multilingual social media content. Resolving these challenges through multilingual training data and domain adaptation techniques is a critical avenue for future research.

The presented model can further be escalated by using the user demographics and specific context for stress detection. In the future, the proposed model can be used by applications with digital therapeutics and conversational agents. This model can be further expanded for multi-label text classification dealing with overlapping emotions.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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KALYANI PENDKE received the M.Tech. degree in computer science and engineering from GHRCE, Nagpur, India, in 2013. She is currently pursuing the Ph.D. degree in computer science with Indian Institute of Information Technology, Nagpur. She is currently an Assistant Professor with the Department of Computer Science and Engineering (Data Science), S. B. Jain Institute of Technology, Management and Research, Nagpur. She has been engaged in teaching for more than 14 years. She has authored around 22 papers in international journals/conferences. Her research interests include natural language processing, machine learning, and deep learning.



MAYURI DIGALWAR received the M.Tech. degree and the Ph.D. degree in computer science from Birla Institute of Technology and Science, Pilani (BITS, Pilani), India. She is currently an Assistant Professor with the Department of Computer Science and Engineering, Indian Institute of Information Technology, Nagpur, India. She has been engaged in teaching for more than 20 years and in research for more than 15 years. She has authored around 20 papers in international/national journals/conferences, published one patent, and a book chapter. Her research interests include text data analytics, machine learning, energy optimization, and sustainable computing.



CHARU GOEL received her Ph.D degree in Mathematics from University of Konstanz, Germany. Prior to that, she got her M.Sc. (Honours School) in Mathematics and B.Sc. (Honours School) in Mathematics from Panjab University, Chandigarh, India. Currently she is working as an Assistant Professor of Mathematics in the Department of Basic Sciences at Indian Institute of Information Technology, Nagpur, India. She has more than 16 years of teaching and research experience at national and international Institutions. Her research interests are in Real Algebraic Geometry (in particular Positive Polynomials and Sum of Squares), Quadraticforms, Polynomial Optimization, and Applications in DataScience and Machine Learning.

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